

Research Monographaskdfjhjk

Book Author

Month Year

Mục lục

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Preface

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Phần I

First half of the book

Chương 1

Introduction

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1.1 Research question

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1.2 Answer to the question

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veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.² Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur.

1.3 Positioning in the literature

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1.4 A section

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1.4.1 A subsection

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1.4.2 Yet another subsection

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1.5 A final introduction section

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Chương 2

Sections and content

Chương 3

Math

Phần II

Second half of the book

Chương 4

Cross-references and quotes

Chương 5

Thiết kế vật lý

5.1 Phân hệ Quản trị, Kiểm duyệt & Minh chứng AI

5.1.1 Cú pháp Rút gọn (Logical Schema Syntax)

- `templates` (template_id UUID PK, workspace_id UUID FK, name VARCHAR(150), description TEXT, content_structure TEXT, creator_type VARCHAR(20), status VARCHAR(20), created_at TIMESTAMP, updated_at TIMESTAMP)
- `template_sections` (section_id UUID PK, template_id UUID FK, heading_title VARCHAR(255), order_index INTEGER, prompt_instruction TEXT, created_at TIMESTAMP, updated_at TIMESTAMP)
- `placeholders` (placeholder_id UUID PK, section_id UUID FK, variable_name VARCHAR(100), expected_data_type VARCHAR(30), description TEXT, is_required BOOLEAN, created_at TIMESTAMP, updated_at TIMESTAMP)
- `template_mappings` (mapping_id UUID PK, placeholder_id UUID FK, code_entity_id UUID FK, extraction_rule TEXT, is_active BOOLEAN)
- `documents` (document_id UUID PK, workspace_id UUID FK, author_user_id UUID FK, title VARCHAR(255), document_type VARCHAR(30),

status VARCHAR(20), created_at TIMESTAMP, updated_at TIMESTAMP, deleted_at TIMESTAMP)

- document_versions (version_id UUID PK, document_id UUID FK, version_number VARCHAR(20), content TEXT, change_summary TEXT, status VARCHAR(20), ticket_codes VARCHAR(500), created_at TIMESTAMP, updated_at TIMESTAMP)
- change_logs (log_id UUID PK, version_id UUID FK, change_type VARCHAR(20), change_description TEXT, diff_content TEXT, created_at TIMESTAMP, updated_at TIMESTAMP)

5.1.2 Mã DDL SQL Hoàn Chỉnh (schema_member_4.sql)

```
1
2
3
4 CREATE TABLE templates (
5     template_id          UUID          NOT NULL DEFAULT
6     gen_random_uuid(),
7     workspace_id         UUID          NOT NULL,
8     name                 VARCHAR(150)  NOT NULL,
9     description          TEXT          NULL,
10    content_structure     TEXT          NULL,
11    creator_type         VARCHAR(20)   NOT NULL,
12    status               VARCHAR(20)   NOT NULL DEFAULT
13    'DRAFT',
14    created_at           TIMESTAMP     NOT NULL DEFAULT
15    CURRENT_TIMESTAMP,
16    updated_at          TIMESTAMP     NOT NULL DEFAULT
17    CURRENT_TIMESTAMP,
18    CONSTRAINT pk_templates PRIMARY KEY (template_id),
19
20    CONSTRAINT uq_templates_workspace_name UNIQUE
21    (workspace_id, name),
22    CONSTRAINT fk_templates_workspace
23    FOREIGN KEY (workspace_id) REFERENCES workspaces
24    (workspace_id)
25    ON UPDATE CASCADE
26    ON DELETE CASCADE,
27    CONSTRAINT ck_templates_creator_type
28    CHECK (creator_type IN ('SYSTEM', 'USER', 'AI')),
29    CONSTRAINT ck_templates_status
30    CHECK (status IN ('DRAFT', 'ACTIVE', 'ARCHIVED')),
31    CONSTRAINT ck_templates_name_not_blank
32    CHECK (length(btrim(name)) > 0),
33    CONSTRAINT ck_templates_updated_after_created
```

```

28         CHECK (updated_at >= created_at)
29 );
30
31 CREATE TABLE template_sections (
32     section_id          UUID          NOT NULL DEFAULT
        gen_random_uuid(),
33     template_id         UUID          NOT NULL,
34     heading_title       VARCHAR(255)  NOT NULL,
35     order_index         INTEGER       NOT NULL,
36     prompt_instruction  TEXT          NULL,
37     created_at          TIMESTAMP     NOT NULL DEFAULT
        CURRENT_TIMESTAMP,
38     updated_at          TIMESTAMP     NOT NULL DEFAULT
        CURRENT_TIMESTAMP,
39     CONSTRAINT pk_template_sections PRIMARY KEY (section_id),
40
41     CONSTRAINT uq_template_sections_order UNIQUE (template_id,
        order_index),
42     CONSTRAINT fk_template_sections_template
43         FOREIGN KEY (template_id) REFERENCES templates
        (template_id)
44         ON UPDATE CASCADE
45         ON DELETE CASCADE,
46     CONSTRAINT ck_template_sections_order_index
47         CHECK (order_index >= 0),
48     CONSTRAINT ck_template_sections_heading_not_blank
49         CHECK (length(btrim(heading_title)) > 0)
50 );
51
52 CREATE TABLE placeholders (
53     placeholder_id      UUID          NOT NULL DEFAULT
        gen_random_uuid(),
54     section_id          UUID          NOT NULL,
55     variable_name       VARCHAR(100)  NOT NULL,
56     expected_data_type  VARCHAR(30)   NOT NULL,
57     description         TEXT          NULL,
58     is_required         BOOLEAN       NOT NULL DEFAULT TRUE,
59     created_at          TIMESTAMP     NOT NULL DEFAULT
        CURRENT_TIMESTAMP,
60     updated_at          TIMESTAMP     NOT NULL DEFAULT
        CURRENT_TIMESTAMP,
61     CONSTRAINT pk_placeholders PRIMARY KEY (placeholder_id),
62
63     CONSTRAINT uq_placeholders_section_variable UNIQUE
        (section_id, variable_name),
64     CONSTRAINT fk_placeholders_section
65         FOREIGN KEY (section_id) REFERENCES template_sections
        (section_id)
66         ON UPDATE CASCADE

```

```

67         ON DELETE CASCADE,
68     CONSTRAINT ck_placeholders_expected_data_type
69         CHECK (expected_data_type IN ('STRING', 'NUMBER',
70         'BOOLEAN',
71                                     'DATE', 'LIST',
72         'OBJECT', 'CODE')),
73     CONSTRAINT ck_placeholders_variable_name_format
74         CHECK (variable_name ~ '^[A-Za-z_][A-Za-z0-9_]*$')
75 );
76 CREATE TABLE template_mappings (
77     mapping_id          UUID          NOT NULL DEFAULT
78     gen_random_uuid(),
79     placeholder_id      UUID          NOT NULL,
80     code_entity_id      UUID          NOT NULL,
81     extraction_rule      TEXT          NULL,
82     is_active           BOOLEAN       NOT NULL DEFAULT TRUE,
83     CONSTRAINT pk_template_mappings PRIMARY KEY (mapping_id),
84     CONSTRAINT uq_template_mappings_pair UNIQUE
85     (placeholder_id, code_entity_id),
86     CONSTRAINT fk_template_mappings_placeholder
87     FOREIGN KEY (placeholder_id) REFERENCES placeholders
88     (placeholder_id)
89     ON UPDATE CASCADE
90     ON DELETE CASCADE,
91     CONSTRAINT fk_template_mappings_code_entity
92     FOREIGN KEY (code_entity_id) REFERENCES code_entities
93     (code_entity_id)
94     ON UPDATE CASCADE
95     ON DELETE CASCADE
96 );
97 CREATE TABLE documents (
98     document_id         UUID          NOT NULL DEFAULT
99     gen_random_uuid(),
100    workspace_id         UUID          NOT NULL,
101    author_user_id       UUID          NOT NULL,
102    title                VARCHAR(255)  NOT NULL,
103    document_type        VARCHAR(30)   NOT NULL,
104    status               VARCHAR(20)   NOT NULL DEFAULT
105    'DRAFT',
106    created_at           TIMESTAMP     NOT NULL DEFAULT
107    CURRENT_TIMESTAMP,
108    updated_at           TIMESTAMP     NOT NULL DEFAULT
109    CURRENT_TIMESTAMP,
110    deleted_at           TIMESTAMP     NULL,
111    CONSTRAINT pk_documents PRIMARY KEY (document_id),

```

```

106     CONSTRAINT fk_documents_workspace
107         FOREIGN KEY (workspace_id) REFERENCES workspaces
108         (workspace_id)
109         ON UPDATE CASCADE
110         ON DELETE CASCADE,
111     CONSTRAINT fk_documents_author
112         FOREIGN KEY (author_user_id) REFERENCES users (user_id)
113         ON UPDATE CASCADE
114         ON DELETE RESTRICT,
115     CONSTRAINT ck_documents_type
116         CHECK (document_type IN ('SRS', 'API_DOC',
117                                 'USER_GUIDE',
118                                 'RELEASE_NOTE',
119                                 'TECH_DESIGN', 'OTHER')),
120     CONSTRAINT ck_documents_status
121         CHECK (status IN ('DRAFT', 'IN_REVIEW', 'PUBLISHED',
122                           'ARCHIVED')),
123     CONSTRAINT ck_documents_title_not_blank
124         CHECK (length(btrim(title)) > 0),
125     CONSTRAINT ck_documents_deleted_after_created
126         CHECK (deleted_at IS NULL OR deleted_at >= created_at)
127 );
128
129 CREATE UNIQUE INDEX uq_documents_workspace_title_active
130 ON documents (workspace_id, title)
131 WHERE deleted_at IS NULL;
132
133 CREATE TABLE document_versions (
134     version_id          UUID          NOT NULL DEFAULT
135     gen_random_uuid(),
136     document_id         UUID          NOT NULL,
137     version_number      VARCHAR(20)   NOT NULL,
138     content             TEXT          NULL,
139     change_summary      TEXT          NULL,
140     status              VARCHAR(20)   NOT NULL DEFAULT
141     'DRAFT',
142     ticket_codes        VARCHAR(500)  NULL,
143     created_at          TIMESTAMP     NOT NULL DEFAULT
144     CURRENT_TIMESTAMP,
145     updated_at          TIMESTAMP     NOT NULL DEFAULT
146     CURRENT_TIMESTAMP,
147     CONSTRAINT pk_document_versions PRIMARY KEY (version_id),
148
149     CONSTRAINT uq_document_versions_number UNIQUE
150     (document_id, version_number),
151     CONSTRAINT fk_document_versions_document
152         FOREIGN KEY (document_id) REFERENCES documents
153         (document_id)
154         ON UPDATE CASCADE

```

```

145         ON DELETE CASCADE,
146     CONSTRAINT ck_document_versions_status
147         CHECK (status IN ('DRAFT', 'IN_REVIEW', 'APPROVED',
148             'REJECTED', 'PUBLISHED')),
149
150     CONSTRAINT ck_document_versions_number_format
151         CHECK (version_number ~ '^[vV]?[0-9]+(\.[0-9]+){0,2}$')
152 );
153
154 CREATE TABLE change_logs (
155     log_id          UUID          NOT NULL DEFAULT
156     gen_random_uuid(),
157     version_id      UUID          NOT NULL,
158     change_type     VARCHAR(20)   NOT NULL,
159     change_description TEXT       NULL,
160     diff_content    TEXT          NULL,
161     created_at      TIMESTAMP     NOT NULL DEFAULT
162     CURRENT_TIMESTAMP,
163     updated_at      TIMESTAMP     NOT NULL DEFAULT
164     CURRENT_TIMESTAMP,
165     CONSTRAINT pk_change_logs PRIMARY KEY (log_id),
166     CONSTRAINT fk_change_logs_version
167         FOREIGN KEY (version_id) REFERENCES document_versions
168         (version_id)
169         ON UPDATE CASCADE
170         ON DELETE CASCADE,
171     CONSTRAINT ck_change_logs_change_type
172         CHECK (change_type IN ('CREATE', 'UPDATE', 'DELETE',
173             'RESTORE', 'PUBLISH'))
174 );
175
176 CREATE INDEX idx_templates_workspace_id ON templates
177     (workspace_id);
178
179 CREATE INDEX idx_templates_status ON templates (status);
180
181 CREATE INDEX idx_template_mappings_code_entity ON
182     template_mappings (code_entity_id);
183
184 CREATE INDEX idx_template_mappings_active ON
185     template_mappings (placeholder_id)
186     WHERE is_active = TRUE;
187
188 CREATE INDEX idx_documents_workspace_id ON documents
189     (workspace_id);
190
191 CREATE INDEX idx_documents_author ON documents
192     (author_user_id);
193
194 CREATE INDEX idx_documents_status ON documents (status);
195
196 CREATE INDEX idx_documents_created_at ON documents
197     (created_at DESC);

```

```

182
183 CREATE INDEX idx_document_versions_status      ON
      document_versions (status);
184 CREATE INDEX idx_document_versions_created_at  ON
      document_versions (document_id, created_at DESC);
185
186 CREATE INDEX idx_change_logs_version_id      ON change_logs
      (version_id);
187 CREATE INDEX idx_change_logs_created_at      ON change_logs
      (created_at DESC);
188
189 CREATE OR REPLACE FUNCTION set_updated_at()
190 RETURNS TRIGGER AS $$
191 BEGIN
192     NEW.updated_at := CURRENT_TIMESTAMP;
193     RETURN NEW;
194 END;
195
196 CREATE TRIGGER trg_templates_updated_at
197     BEFORE UPDATE ON templates
198     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
199 CREATE TRIGGER trg_template_sections_updated_at
200     BEFORE UPDATE ON template_sections
201     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
202 CREATE TRIGGER trg_placeholders_updated_at
203     BEFORE UPDATE ON placeholders
204     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
205 CREATE TRIGGER trg_documents_updated_at
206     BEFORE UPDATE ON documents
207     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
208 CREATE TRIGGER trg_document_versions_updated_at
209     BEFORE UPDATE ON document_versions
210     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
211 CREATE TRIGGER trg_change_logs_updated_at
212     BEFORE UPDATE ON change_logs
213     FOR EACH ROW EXECUTE FUNCTION set_updated_at();
214 GO

```


Chương 6

Extensions

Phần III

Conclusion

Chương 7

Summary

Chương 8

Future research

Phần IV

Technical appendix

Phụ lục A

Generic technical topic

Phụ lục B

Another technical topic

Phụ lục C

Last technical topic

This is a last appendix with a bit of text.

C.1 A first section

Sed vestibulum ex a tristique lacinia. Integer interdum magna vel magna rutrum fermentum. Maecenas sed mi in nunc convallis rutrum. Vivamus dapibus bibendum est, ac tincidunt ipsum fermentum id. Nunc id est turpis. Suspendisse potenti. Sed ac laoreet nulla, eu sollicitudin libero. Suspendisse tempus orci nec mauris volutpat eleifend. Integer nec tristique libero. Sed vehicula ipsum sit amet magna accumsan, nec pulvinar nisl consequat. In ac hendrerit turpis. Sed aliquet luctus mauris, vitae congue turpis. Nullam blandit lacus vel interdum eleifend.

C.2 A second section

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$$S^*(z) = \frac{\alpha}{1 - \gamma(1 - \alpha)} > \int \frac{\alpha(i)}{\gamma(i)} di. \quad (\text{C.1})$$

Nam pretium mauris eros, nec sollicitudin risus tincidunt a. Vivamus in augue vitae ligula scelerisque dapibus. Integer eget metus aliquet, efficitur

nibh eget, pharetra eros. Suspendisse luctus interdum ex id suscipit. Donec quis augue mauris. Nulla ut erat eget nisl hendrerit malesuada. Vivamus eu tortor sit amet sem fringilla eleifend. In hac habitasse platea dictumst. Suspendisse potenti. Ut eget libero sed orci tempor ullamcorper:

$$\mathcal{W}^2(\zeta) = \left[\frac{\beta}{1 - \nu(1 - \xi)} \right] \pm \Lambda \quad (\text{C.2})$$

$$\mathcal{L}^2(\zeta) = \left(\frac{\beta}{1 - \nu(1 - \xi)} \right) \quad (\text{C.3})$$

$$\overline{a - b} = \int_0^{+\infty} \frac{\beta}{1 - \nu(x)(1 - \xi)} dx \in \mathbb{R}. \quad (\text{C.4})$$

C.3 A third section

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Tài liệu tham khảo

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Nemo enim ipsam voluptatem, quia voluptas sit, aspernatur aut odit aut fugit, sed quia consequuntur magni dolores eos, qui ratione voluptatem sequi nesciunt, neque porro quisquam est, qui dolorem ipsum, quia dolor sit amet consectetur adipiscing velit, sed quia non numquam eius modi tempora incidunt, ut labore et dolore magnam aliquam quaerat voluptatem. At vero eos et accusamus et iusto odio dignissimos ducimus, qui blanditiis praesentium voluptatum deleniti atque corrupti, quos dolores et quas molestias excepturi sint, obcaecati cupiditate non provident, similique sunt in culpa, qui officia deserunt mollitia animi, id est laborum et dolorum fuga.